CarClinch Dealership Assistant

Final Project Presentation

Group 4
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Agenda

01 Project Overview

02 Architecture & Tech Stack

03 Core Features

04 Database & AI Integration

05 Infrastructure & Deployment

06 Testing & Risk Management

07 Future Recommendations

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PROJECT OVERVIEW

Problem → Goals

Problem

- Dealerships rely on **manual** email communication efforts
- Slow response times → **leads go cold**
- Manual efforts → inefficient responses or follow-up sequences
- **Lost leads** for test drive bookings due to inconsistent communication

Goal: Dealership Assistant POC

- **Automate** lead intake → AI email response
- **GPT-4.1 mini** for intent, sentiment & context-aware replies
- **Follow-up sequences** for unresponsive leads
- **Appointment booking** w/ real-time slot availability

PROJECT OVERVIEW

Market & Stakeholders

Ontario Automotive Market

43,000+ dealerships in Ontario
99.8% are small-medium businesses **(SMB)**
Target: SMB dealerships **lacking automated lead tools**

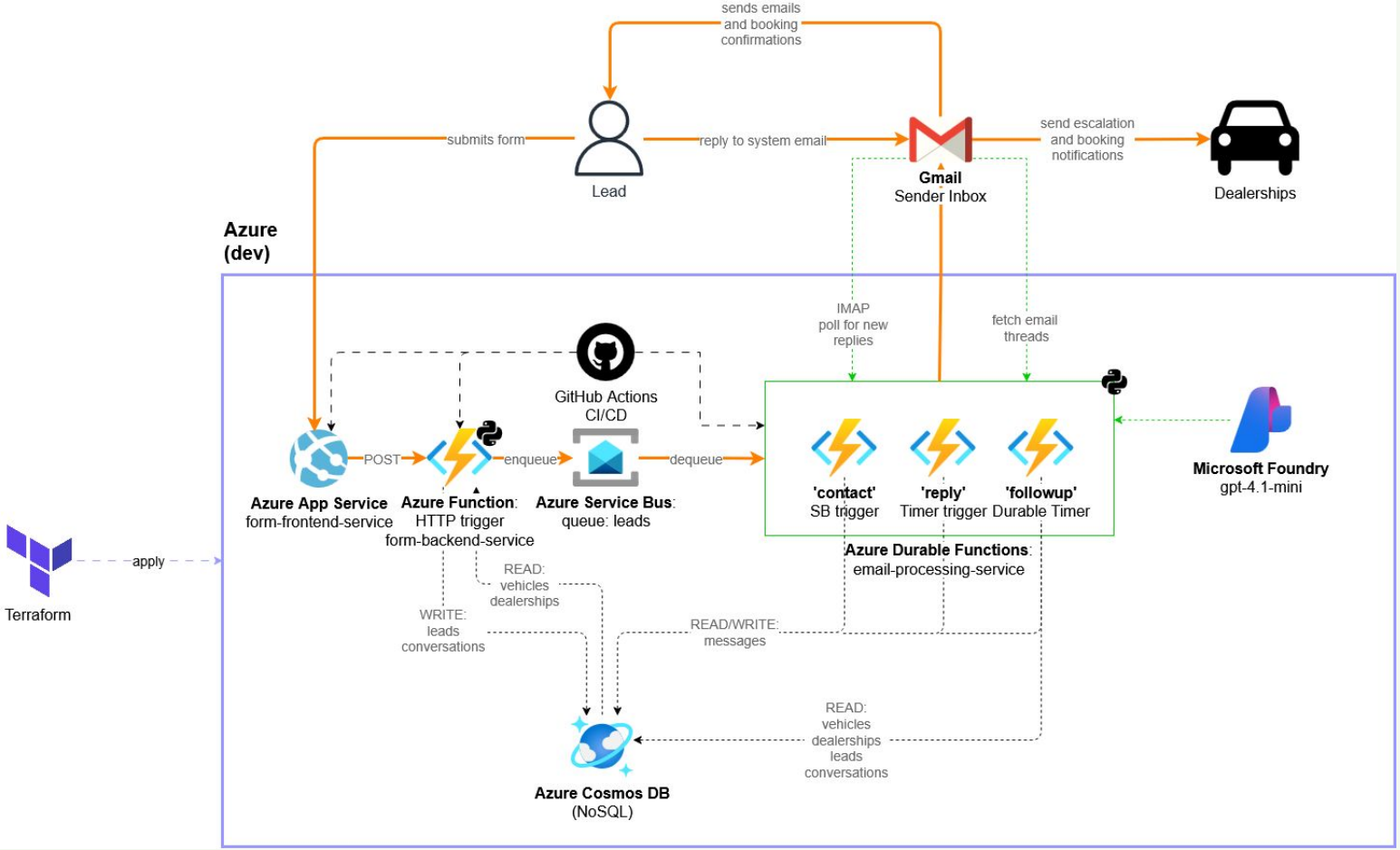
Digital Shift

Buyers **research online** before visiting a showroom (McKinsey 2020)
Record-high satisfaction when **digital + in-store** are blended (Cox 2025)

Key Stakeholders

Stakeholder	Role	Influence	Authority
Ahmed Al-Asaad (CarClinch)	Project Owner/Sponsor	High	Approves scope & MVP direction
Ahmad Ahmourah (CarClinch)	Technical Advisor	Med/High	Provides guidance, no dev role
Dealerships	End-Users (indirect)	Low	Needs captured via BRD
Project Team	Architects & Developers	Med/High	Full technical authority

ARCHITECTURE



ARCHITECTURE

Azure Services

Azure (Durable) Functions

Stateful orchestration in Python.
Consumption model with free-tier.

Azure Service Bus

Ensures reliable delivery from form to email processing service.

Azure Cosmos DB (NoSQL)

Schema-less, semi-structured flexibility for development of POC.

Microsoft Foundry (GPT-4.1 mini)

Data stays in Azure compliance boundary.
Ample model options to switch to in future integration.

Azure Static Web App

Free tier with GitHub Actions CI/CD.
No App Service overhead for a static form.

Azure Key Vault

Audit logs, access policies, Managed Identity retrieval and management via Terraform.

CORE FEATURES

E2E System Features

Lead Intake & Validation

HTML form → Azure Function API → validates email/phone → publishes to Service Bus queue.

Automated Outreach Email

Durable Function orchestrator sends SMTP email within 1 minute
GPT-4.1 mini generates the message body from lead & vehicle context.

IMAP Reply Polling

1-minute Cron timer polls Gmail inbox for UNSEEN messages.
Associates reply w/ CosmosDB conversation by matching conversationId in Message-ID headers.

AI/LLM: Dual Layer

Two-layer LLM pipeline:
(1) intent/sentiment classification → structured JSON,
(2) context-aware response generation with confidence-gated escalation.

Appointment Booking

Multi-step booking flow:
request_date → request_time → confirm_booking
HTML table slot suggestions injected via placeholders.
.ics calendar invites on confirm.

Follow-up Automation

Durable Timers schedule 3 follow-up sequences.
Automatically cancelled when lead replies or when car is sold.

CosmosDB (NoSQL) Schema

Container	Partition Key	Purpose
dealerships	/id	Dealership profiles, contact & address info
vehicles	/dealerId	Vehicle inventory; grouped by dealership for efficient queries
leads	/email	Lead profiles, notes array, status lifecycle (new → booked)
conversations	/leadId	Active/inactive email threads; links lead, vehicle, dealer
messages	/conversationId	Individual messages; stores email body, role, responseId for LLM chaining
appointments	/dealerId	Booked test-drive slots; prevents double-booking by date+timeslot

AI Processing Pipeline

Layer 1: Analysis

ANALYSIS_SYSTEM_PROMPT + ANALYSIS_USER_PROMPT
→ Classifies: **intentCategory, intentAction, sentiment, urgency**
→ Extracts structured dates/times (ISO 8601)
→ Outputs sanitized JSON payload
→ Malformed JSON → auto-escalate (fail-safe)

Layer 2: Response Generation

SYSTEM_PROMPT + CONTACT/REPLY/FOLLOWUP_USER_PROMPT
→ Human-like sales assistant persona
→ Restricted to vehicle info + appointment booking
→ Chained via prior responseID (no re-sending history)
→ Max 150 words

Appointment Booking Logic

request_date	request_date_range	request_time	confirm_booking
Wants to book, no date given	Relative range (e.g. 'next week') → max 7 ISO dates	Date given, no time → preferredTimeRange [9, 17]	Exact date + hour → writes appointment, sends .ics

Deployment Environments

Local: Docker Compose

- Azurite (Storage Account) emulator
- Azure Service Bus emulator
- Azure Cosmos DB emulator
- form-frontend-service (HTML/CSS/JS)
- form-backend-service (Function)
- email-processing-service (Durable Function)

Azure: Terraform

- **terraform init/plan/apply** → full provision
- **azurerm** + **github** providers
- Managed Identity
- Key Vault for OpenAI API key (KV reference)
- Cosmos DB, Service Bus, 2× Function Apps
- Static Web App + GitHub Actions CI/CD

Security Model: Least-privilege Managed Identity across Azure services · No connection strings in source code · TLS on all SMTP/IMAP · HTTPS on all endpoints · CosmosDB via DefaultAzureCredential

Testing Strategy & Key Risks

Testing Coverage (pytest + Jest)

- Unit tests across all services
- Mocked all Azure & system resources: CosmosDB, GPT-4.1 mini, Service Bus, SMTP
- Prompt hydration, JSON fallback, form validation
- **Gap:** E2E integration tests & LLM prompt drift

Top Risks & Mitigations

- LLM **hallucination** → confidence gates + escalation
- **Prompt injection** → strict system prompt guardrails
- Email **deliverability** → SMTP retry logic
- Function **cold starts** → accepted SLA (< 1 min email)
- **CASL** compliance → opt-out in all followup emails

Scope & Constraints Summary

- **In-scope:** lead intake, AI email pipeline, follow-ups, appointment booking, Terraform IaC, full documentation
- **Out-of-scope** (deferred): AI phone assistant, Power BI analytics dashboard, full CRM integration, staff takeover portal
- **Budget: ~$100–$140 CAD** per team member via Azure student credits; All local dev at $0 using Docker emulators

FUTURE RECS.

Roadmap Beyond POC

SMTP → ACS or Microsoft Graph API

- **ACS** provides managed IP reputation, SPF/DKIM automation, bounce handling & delivery analytics.
- **MS Graph** leverages existing M365 licensing, inherits DLP policies, and puts sent items in the dealership's Sent folder.

Power BI Reporting & Analytics

- New CosmosDB **'events' container** (partitioned by /dealerId)
- Capture defined **events** for KPIs, with varied metadata
- **KPIs:** lead response time, email response rate, booking rate, AI vs human ratio.

Agentic AI Architecture

- Rearchitect from **prompt-based to agentic LLM design.**
- **Tool-calling** capabilities (fetch availability, write appointment, trigger escalation)
- **Reduces prompt engineering overhead.**

AI Phone Assistant

- Microsoft Foundry **real-time voice agent**
- Handles FAQ calls, books appointments, escalates complex calls to staff.

Live Demo!

Thank You!

Any Questions?

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